Troubleshooting Guide - Indonesia ORC Sites

Indonesia ORC Deployment — Spring 2026

American Red Cross / Palang Merah Indonesia

Version 3.2 | Audience: Field technicians, PMI staff | Built 2026-04-09

Contents

Troubleshooting Guide - Indonesia ORC Sites **2**
Contents . . . 2
Quick Reference - Status LEDs . . . 2
Diagnostic Flowcharts . . . 2
Camera Network Setup . . . 11
Common Issues & Solutions . . . 12
Command Reference . . . 15
Maintenance Mode . . . 19
When to Escalate . . . 20
Preventive Maintenance Schedule . . . 20

Troubleshooting Guide - Indonesia ORC Sites

Applies to: Sukabumi and Jakarta sites **Audience:** Field technicians, PMI staff

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Contents

- Quick Reference - Status LEDs
- Diagnostic Flowcharts
- Camera Network Setup
- Common Issues & Solutions
- Command Reference
- Maintenance Mode
- When to Escalate
- Preventive Maintenance Schedule

Quick Reference - Status LEDs

LED Pattern	Meaning	Action
Green steady	System OK, idle	Normal operation
Green + Yellow blink	Capturing/uploading	Normal operation
Yellow steady	Processing	Wait for completion
Red blink	Capture failed	Check camera connection and logs
Red steady	Camera offline/unreachable	**Normal on Sukabumi between cycles** (camera powers down with PoE relay). On Jakarta, investigate if persistent — camera should be reachable 24/7
All off	No power or not booted	Check power system
All on steady	Boot in progress	Wait 2-3 minutes

Diagnostic Flowcharts

System Will Not Power On

```
START: No status LEDs lit
         |
         ▼
┌──────────────────────────────┐
| Press external power button    |
| (J2 header, near USB-C port)  |
```

```
│ — brief press to power on     │
└─────────────────────────────┘
                │
                │
        ┌───────┴───────┐
        │               │
      Boots           No boot
        │               │
        ▼               ▼
    ┌─────────┐    Check power source
    │System   │
    │is       │
    │working. │    ┌─────────────────┐
    └─────────┘    │ Sukabumi: Solar │
                   │ Jakarta: AC     │
                   └─────────┬───────┘
                             │
                             ▼
                ┌──────────────────────────┐
                │ Measure voltage at input  │
                │ terminals (should be >11V) │
                └──────────────┬───────────┘
               │
        ┌──────┴──────┐
        │             │
    <11V           >11V
        │             │
        ▼             ▼
┌──────────┐   ┌──────────────────┐
│ SUKABUMI │   │ Check fuses      │
│ Check    │   │ (main, Pi feed)  │
│ solar    │   └────────┬─────────┘
│ battery  │            │
└──────────┘      ┌─────┴─────┐
                  │           │
               Blown       OK
                  │           │
                  ▼           ▼
            Replace     ┌──────────────────────┐
            fuse        │ Check power path:    │
                        │ SUKABUMI: DDR-60G-5  │
                        │   5V output          │
                        │ JAKARTA: Witty Pi    │
                        │   VIN (12V from F3)  │
                        └──────────┬───────────┘
                                   │
                             ┌─────┴─────┐
                             │           │
                           Off         On
```

```
        |          |
        ▼          ▼
┌──────────┐  ┌──────────────┐
| Check    |  | Check Pi     |
| USB-C    |  | activity LED |
| power    |  | (should      |
| cable    |  | flash)       |
└──────────┘  └──────────────┘
```

```
NOTE: The external power button is wired to the Pi 5 J2
header (2-pin, near USB-C port). This is a dedicated hardware
power control — NOT a GPIO pin. If Pi has power but will not
boot, try pressing the J2 button before investigating further.
```

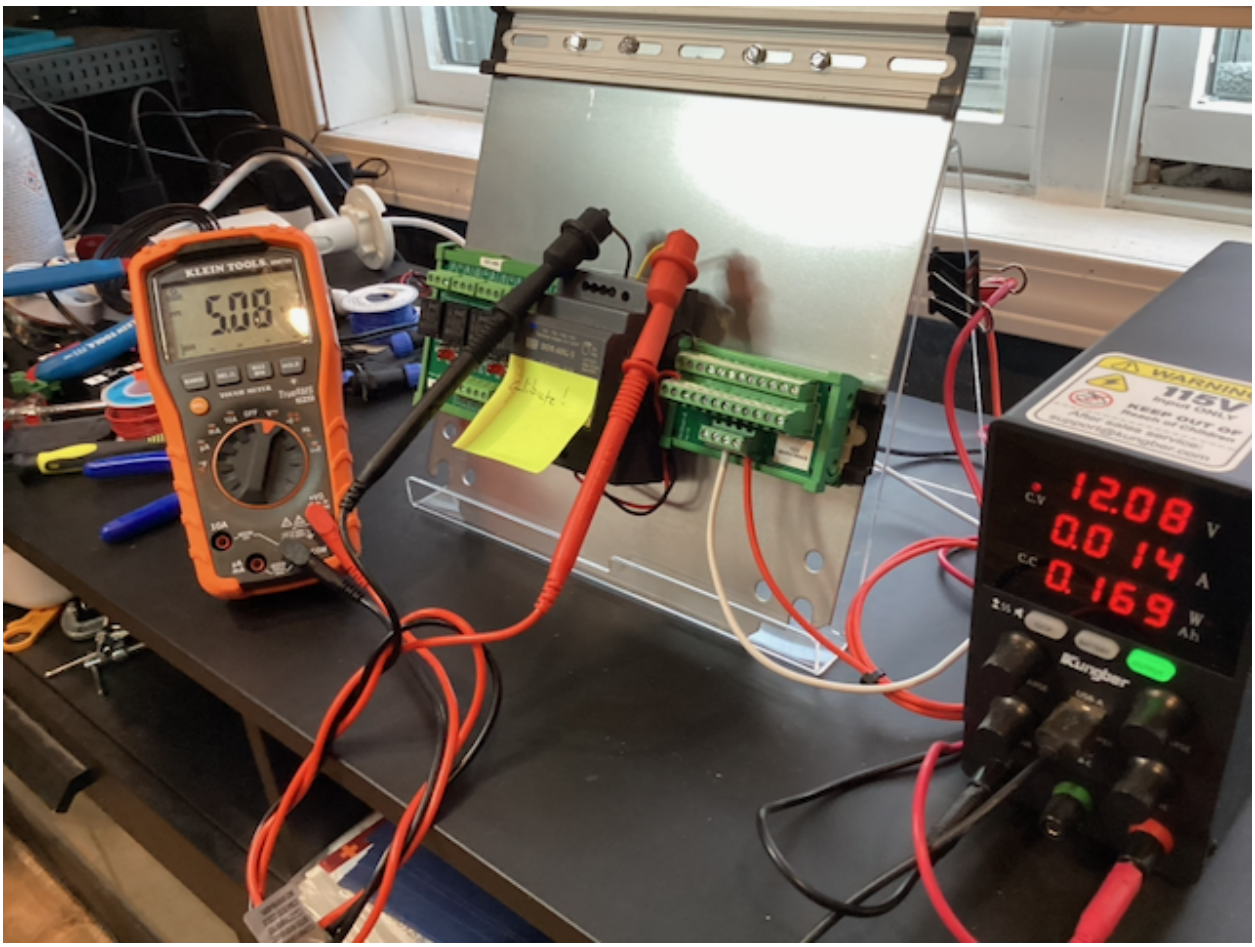

Figure 1: Multimeter reading 5.08V at buck converter output, bench supply showing 12V input

Figure: Checking 5V output with a multimeter. For Sukabumi, measure at the DDR-60G-5 output. For Jakarta, measure at any G469 5V pin (Witty Pi delivers 5V through the header).

Pi Frozen / Unresponsive

```
START: Pi not responding (SSH hangs, no LED activity)
         |
         ▼
┌──────────────────────────────────────┐
| Try SSH or ping from maintenance     |
| laptop (if connected)                |
└──────────────────────────────────────┘
               |
               |
        ┌──────┴──────┐
        |             |
    Responds       No response
        |             |
        ▼             ▼
  ┌──────────┐  ┌──────────────────────────────┐
  | Not      |  | Press external power button (J2)  |
  | frozen — |  | briefly — initiates clean shutdown |
  | check    |  └──────────────┬───────────────┘
  | services |                 |
  └──────────┘          ┌──────┴──────┐
                        |             |
                    Shuts down     No response
                        |             |
                        ▼             ▼
                  ┌──────────┐  ┌──────────────────────┐
                  | Press J2 |  | HOLD power button ~10s   |
                  | again to |  | to force power off       |
                  | power on |  | Then press J2 to boot    |
                  └──────────┘  └──────────────────────┘

NOTE: The J2 power button is near the USB-C port on the Pi 5.
- Brief press on halted Pi → powers on
- Brief press on running Pi → clean shutdown
- Long press (3s) on running Pi → enters maintenance mode
- Hold ~10s → force power off (use only if Pi is frozen)
```

Pi eth0 Has Only IPv6 Address (No IPv4)

If `ip addr` shows only an `inet6` (IPv6) address on eth0 with no `inet` (IPv4) address, the Pi cannot reach the camera.

Symptoms: - `ping <camera-ip>` fails with "Network is unreachable" - `ip addr show eth0` shows `fe80::...` link-local IPv6 but no `192.168.x.x`

Cause: The Pi Ethernet interface has not been assigned a static IPv4 address and dnsmasq has not been configured yet. Without a static IP, the Pi cannot serve DHCP to cameras on the 192.168.50.0/24 network.

Fix — assign a static IPv4 address and set up dnsmasq:

Using NetworkManager (Raspberry Pi OS Bookworm):

```
# Set static IP on eth0
sudo nmcli con mod "Wired connection 1" ipv4.addresses 192.168.50.1/24
sudo nmcli con mod "Wired connection 1" ipv4.method manual
sudo nmcli con up "Wired connection 1"

# Verify
ip addr show eth0   # should now show 192.168.50.1
```

If the connection name differs, find it with:

```
nmcli con show
```

Then install and configure dnsmasq as DHCP server for cameras — see the assembly guide for your site (ASSEMBLY_SUKABUMI.md or ASSEMBLY_JAKARTA.md) for the full dnsmasq configuration.

This persists across reboots. The Pi acts as DHCP server on the camera network (192.168.50.0/24), assigning predictable IPs to each camera via MAC address reservation.

Camera Not Working

Sukabumi (PoE Camera - Power-Cycled)

```
START: No video capture
         |
         ▼
┌──────────────────────────┐
| Ping camera from Pi:     |
| ping 192.168.50.100      |
| (or configured IP)       |
└──────────────────────────┘
            |
            |
     ┌──────┴──────┐
     |             |
  No reply      Reply OK
     |             |
     ▼             ▼
┌──────────────┐  ┌────────────────────┐
| Check PoE    |  | Test RTSP capture: |
| switch:      |  | orc-capture        |
| - 12V input? |  |   --skip-relay     |
| - LED on?    |  |   --dry-run        |
| - Relay GPIO |  |                    |
|   wiring?    |  |                    |
└──────┬───────┘  └──────────┬─────────┘
       |                     |
  ┌────┴────┐          ┌────┴────┐
  |         |          |         |
```

```
No LED    LED on    Quality   Quality
  |         |       check     check
  |         |       FAIL      PASS
  |         |         |           |
  ▼         ▼         ▼           ▼
┌─────┐  ┌─────────┐ ┌───────┐  ┌────────┐
│Check│  │Check    │ │Check  │  │Camera  │
│12V  │  │Cat6     │ │RTSP   │  │OK,     │
│fuse │  │cable &  │ │stream │  │check   │
│& PoE│  │CNLINKO  │ │config │  │ORC     │
│sw   │  │connect- │ │& cam  │  │config  │
│power│  │ions     │ │creds  │  └────────┘
└─────┘  └─────────┘ └───────┘

NOTE: Camera takes ~45-60s to boot after Pi wakes.
Wait before testing if system just powered on.
```

Jakarta (PoE Cameras)

```
START: Camera offline
         |
         ▼
┌──────────────────────┐
| Ping camera from Pi:    |
| ping 192.168.50.100     |
└──────────────────────┘
            |
            |
     ┌──────┴──────┐
     |             |
  No reply      Reply OK
     |             |
     ▼             ▼
┌──────────────┐  ┌──────────────────┐
| Check PoE    |  | Test RTSP capture: |
| switch LED   |  | orc-capture        |
| and camera   |  |   --skip-relay     |
| power LED    |  |   --dry-run        |
└──────────────┘  |                    |
       |          └──────────┬─────────┘
       |                     |
  ┌────┴────┐          ┌────┴────┐
  |         |          |         |
No LED    LED on      |         |
  |         |       Quality   Quality
  |         |       check     check
  |         |       FAIL      PASS
  ▼         ▼         |           |
┌─────┐  ┌─────────┐  ▼           ▼
│Check│  │Check    │ ┌───────┐  ┌────────┐
```

```
|12V  |  |Cat6    | |Check  |  |Camera  |
|to   |  |cable   | |RTSP   |  |OK,     |
|PoE  |  |& bulk- | |stream |  |check   |
|sw & |  |head    | |config |  |ORC     |
|relay|  |connect-| |& cam  |  |config  |
└────┘   |ions    | |creds  |  └────────┘
         └────────┘ └──────┘
```

No LTE Connectivity

```
START: No data connection
         |
         ▼
┌──────────────────────────┐
| Check modem detected:    |
| mmcli -L                 |
└──────────────────────────┘
            |
            |
     ┌──────┴──────┐
     |             |
  No modem     Modem found
     |             |
     ▼             ▼
┌──────────┐  ┌──────────────────┐
| Check    |  | Check SIM:       |
| USB      |  | mmcli -m 0       |
| connection| | Look for SIM state |
| Run:     |  └──────────┬───────┘
| lsusb    |             |
└──────────┘       ┌─────┴─────┐
                   |           |
               No SIM       SIM OK
                   |           |
                   ▼           ▼
            ┌──────────┐  ┌──────────────┐
            | Check    |  | Check signal: |
            | SIM      |  | mmcli -m 0    |
            | inserted |  | --signal-get  |
            | properly |  └──────┬───────┘
            └──────────┘         |
                          ┌──────┴──────┐
                          |             |
                       No signal    Has signal
                          |             |
                          ▼             ▼
                     ┌──────────┐  ┌────────────┐
                     | Check    |  | Check APN: |
                     | antenna  |  | May need   |
```

```
                    | connect- |  | Telkomsel  |
                    | ions     |  | APN config |
                    └──────────┘  └────────────┘
```

Camera IR Not Working

```
START: No IR illumination at night
         |
         ▼
┌──────────────────────────────┐
| Cover camera lens with hand |
| or cloth (simulate dark)    |
└──────────────┬───────────────┘
               |
        ┌──────┴──────┐
        |             |
  IR off        IR on
        |             |
        ▼             ▼
┌──────────────┐  ┌────────────┐
| Check camera |  | IR working |
| IR settings  |  | Check      |
| in web UI    |  | light      |
| at           |  | threshold  |
| camera-ip    |  | adjustment |
└──────┬───────┘  └────────────┘
       |
   ┌───┴───┐
   |       |
IR on in  IR disabled
settings     |
  |          ▼
  ▼      ┌──────────┐
┌─────┐  | Enable   |
|Check|  | IR auto  |
|PoE  |  | mode in  |
|power|  | camera   |
|to   |  | web UI   |
|cam  |  └──────────┘
└─────┘

NOTES:
- ANNKE C1200 has built-in IR LEDs controlled by internal photocell
- IR settings are in the camera web interface, not on the Pi
- Both sites use the same ANNKE C1200 with built-in IR
```

Rain Gauge Not Reporting

```
START: No rain data
         |
         ▼
┌──────────────────────────┐
| Check UART device:       |
| ls /dev/ttyAMA0          |
| ls /dev/serial0          |
└──────────────────────────┘
            |
            |
     ┌──────┴──────┐
     |             |
  Not found     Found
     |             |
     ▼             ▼
┌──────────────┐  ┌────────────────┐
| Check wiring |  | Test serial:   |
| VCC: 12V     |  | cat /dev/      |
| GND: Ground  |  | ttyAMA0        |
| TX: GPIO 15  |  | (tip bucket,   |
| RX: GPIO 14  |  | watch output)  |
└──────────────┘  └────────┬───────┘
                           |
                     ┌─────┴─────┐
                     |           |
                  No data     Data OK
                     |           |
                     ▼           ▼
                ┌──────────┐  ┌──────────────┐
                | Check    |  | Check ORC    |
                | 12V pwr  |  | rain gauge   |
                | to RG-15 |  | config       |
                | TX/RX    |  |              |
                | wiring   |  └──────────────┘
                └──────────┘
```

Sukabumi — Rain Data During Power Cycling:

The RG-15 stays powered from TB1 (always-on 12V bus) while the Pi sleeps. It accumulates rainfall internally. The capture script reads the accumulated total on each wake cycle and computes interval rainfall by delta.

- If readings show zero when it clearly rained: check that RG-15 VCC is on TB1 (always-on), NOT on the relay-switched 12V circuit
- If accumulated total keeps resetting: RG-15 is losing power — check TB1 12V with multimeter while Pi is sleeping
- If readings jump unexpectedly: the previous reading file on disk may be corrupt or missing — check `/var/lib/orc/rain_last_acc.txt` (or wherever the capture script stores the last-read value)

Camera Network Setup

Both sites use the Pi as a DHCP server (dnsmasq) on the 192.168.50.0/24 camera network. Cameras receive predictable IPs via MAC address reservation. See the assembly guides for full setup instructions.

Site	Pi IP	Camera IPs
Sukabumi	192.168.50.1	192.168.50.100
Jakarta	192.168.50.1	192.168.50.100

Note: The SADP utility (Hikvision/ANNKE) does not run on ARM Macs — neither natively nor under Parallels. The dnsmasq approach eliminates the need for SADP entirely.

Note: The ANNKE web interface requires a Windows-only browser plugin for live view. Use ISAPI snapshot to verify the camera image instead.

dnsmasq failed at boot (camera unreachable)

Symptoms: - Camera has PoE power and link light, but `ping <camera-ip>` fails - `sudo systemctl status dnsmasq` shows `failed` with `unknown interface eth0`

Cause: dnsmasq started before the PoE relay powered eth0. With `bind-interfaces`, dnsmasq fails immediately if the interface has no carrier. The camera never gets a DHCP lease.

Immediate fix:

```
sudo systemctl restart dnsmasq
# Wait 60-90s for camera to get DHCP lease, then:
ping 192.168.50.100
```

Permanent fix: Change `bind-interfaces` to `bind-dynamic` in `/etc/dnsmasq.d/maintenance.conf`. With `bind-dynamic`, dnsmasq starts successfully even if eth0 is down, and begins serving DHCP as soon as the interface appears. See `pi/shared/etc/dnsmasq.d/maintenance.conf` for the correct config.

Camera not getting expected IP

1. Check dnsmasq leases: `cat /var/lib/misc/dnsmasq.leases`
2. Verify the camera's MAC address in `/etc/dnsmasq.conf` matches the label on the camera
3. Restart dnsmasq: `sudo systemctl restart dnsmasq`
4. Power-cycle the camera (unplug/replug PoE cable or cycle relay) and wait 60-90s

Finding camera MAC address

If you do not have the MAC address yet:

1. Temporarily remove any `dhcp-host=` lines from `/etc/dnsmasq.conf`
2. Restart dnsmasq: `sudo systemctl restart dnsmasq`
3. Connect camera to PoE switch, wait 60-90s for boot
4. Check leases: `cat /var/lib/misc/dnsmasq.leases`

5. The camera will have received an IP from the 192.168.50.100-200 range
6. Note the MAC address, add it to `/etc/dnsmasq.conf` as a `dhcp-host=` line
7. Restart dnsmasq and power-cycle the camera

WiFi subnet conflict (laptop direct access)

The ANNKE C1200 ships with a default IP on 192.168.1.x — the same subnet used by most WiFi routers. If you ever need to access a camera directly from a laptop (before dnsmasq is set up), disable WiFi first:

```
# macOS:
networksetup -setairportpower en0 off
# Linux:
nmcli radio wifi off
```

Then set your laptop to 192.168.1.50/24 and use `arp -a` to find the camera. Re-enable WiFi when done.

Common Issues & Solutions

Power Issues

Problem	Possible Cause	Solution
System will not boot	Pi halted but has power	Press external power button (J2 header, near USB-C)
System will not boot	Dead battery (Sukabumi)	Charge battery, check solar panel
System will not boot	Blown fuse	Replace fuse, investigate cause — see fuse photo below
System will not boot	Faulty USB-C cable	Replace power cable to Pi
System will not boot	Witty Pi VIN not receiving 12V (Jakarta)	Check F3 fuse, measure 12V at Witty Pi VIN screw terminal
Pi frozen/unresponsive	Software hang	Press J2 power button briefly for clean shutdown, then press again to boot
Pi frozen/unresponsive	Hard lock	Hold J2 power button ~10s to force off, then press to boot
Intermittent shutdowns (Sukabumi)	Low battery voltage	Check solar charge controller
Intermittent shutdowns (Jakarta)	BMS low-voltage cutoff	Check charger and battery health
Intermittent shutdowns	Loose connection	Check all terminal connections

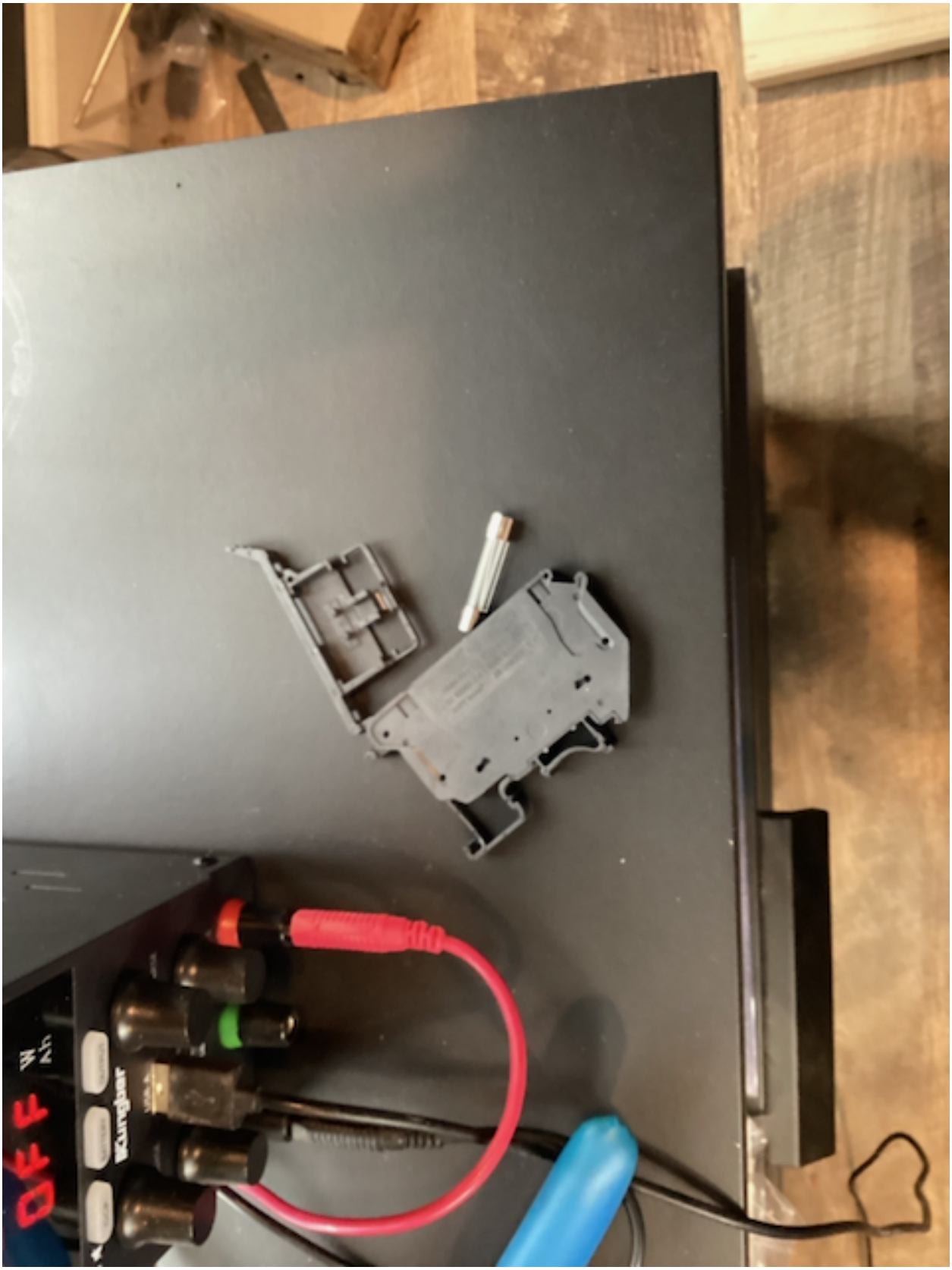

Figure 2: Fuse holder with wrong-size Bussmann automotive fuses that do not fit the 5x20mm holder

Figure: Wrong fuse type. These Bussmann automotive blade fuses (6.3mm) do NOT fit the 5x20mm glass tube holders used in the stations. Use 5x20mm slow-blow glass fuses only.

Camera Issues (Both Sites Use PoE Cameras)

Problem	Possible Cause	Solution
PoE camera offline	PoE switch fault	Check switch power/data LEDs
PoE camera offline	Relay not energized	Check relay GPIO wiring (VCC, GND, IN1-IN4)
PoE camera offline	Cat6 cable fault	Test cable continuity
PoE camera offline	12V power to switch	Check fuse, terminal connections
PoE camera offline (Sukabumi)	Camera still booting	Wait 60s after Pi wakes
PoE camera not reachable	dnsmasq failed at boot	`sudo systemctl restart dnsmasq`, wait 60-90s — see Camera Network Setup
PoE camera not reachable	Wrong IP address	Verify camera DHCP settings
PoE camera not reachable	Network config	Check Pi and camera on same subnet
PoE camera not reachable	Pi eth0 IPv6-only	Pi has no IPv4 address — see "Pi eth0 Has Only IPv6 Address" above
Camera image blurry	Focus issue	Adjust lens focus ring
Camera image dark	Exposure settings	Adjust via camera web interface
Camera image washed out	Overexposure	Reduce exposure in settings

Connectivity Issues

Problem	Possible Cause	Solution
No LTE signal	Antenna disconnected	Check puck antenna connections
No LTE signal	Wrong bands	Verify EG25-G band config
No LTE data	APN not configured	Set Telkomsel APN
No LTE data	IMEI not registered	Register with POSTEL
No LTE data	SIM deactivated	Check with carrier
Slow upload	Poor signal	Relocate antenna, check puck position

Environmental Issues

Problem	Possible Cause	Solution
Condensation inside enclosure	Gore vent (a waterproof air vent that allows air pressure to balance) blocked	Clear vent obstruction
Condensation inside enclosure	Failed seal	Reseal bulkheads/glands
Condensation inside enclosure	PTC heater not installed (Jakarta)	Planned for future – use desiccant packs as interim measure or install PTC heater
Corrosion on connectors	Missing dielectric grease	Clean, apply grease
High internal temp	Direct sun exposure	Add shade, check fans (Jakarta)
Water ingress	Failed bulkhead seal	Replace bulkhead, reseal

Command Reference

System Status

```
# Run preflight checks (packages, configs, services, hardware, network)
orc-preflight

# Check and fix failures (prompts before applying)
cd ~/code/git/openrivercam/spring_2026_ID/pi && bash deploy.sh <site>

# Check system uptime
uptime

# Check CPU temperature
vcgencmd measure_temp

# Check disk usage
df -h

# Check memory
free -h

# Check running services
systemctl status orc
```

USB Devices

```
# List USB devices
lsusb

# Check USB flash drive
lsblk
```

```
# List video devices
ls /dev/video*

# Test camera capture (USB)
ffmpeg -f v4l2 -i /dev/video0 -frames:v 1 /tmp/test.jpg
```

LTE Modem

```
# List modems
mmcli -L

# Modem status
mmcli -m 0

# Signal strength
mmcli -m 0 --signal-get

# Check connection
ping -c 3 8.8.8.8

# Restart modem
mmcli -m 0 --disable && sleep 5 && mmcli -m 0 --enable
```

Network (PoE cameras)

```
# Ping cameras (Sukabumi)
ping -c 3 192.168.50.100

# Ping camera (Jakarta)
ping -c 3 192.168.50.100

# Test camera connectivity with ISAPI snapshot
curl --digest -u admin:PASSWORD http://192.168.50.100/ISAPI/Streaming/channels/101/picture
 ↪  -o /tmp/cam1.jpg

# Test RTSP capture
orc-capture --skip-relay --dry-run

# Check DHCP leases
cat /var/lib/misc/dnsmasq.leases

# Check network interfaces
ip addr

# Check route table
ip route
```

```
# Check dnsmasq status
sudo systemctl status dnsmasq
```

UART (Rain gauge)

```
# Check UART device exists
ls /dev/ttyAMA0 /dev/serial0

# Read rain gauge output (Ctrl+C to stop)
cat /dev/ttyAMA0

# Configure serial port (if needed)
stty -F /dev/ttyAMA0 9600 cs8 -cstopb -parenb
```

I2C (SHT40 sensor, Witty Pi 5 RTC)

```
# Scan I2C bus (SHT40 should appear at 0x44, Witty Pi 5 at 0x51)
i2cdetect -y 1
```

1-Wire (DS18B20 probe)

```
# List 1-Wire devices
ls /sys/bus/w1/devices/

# Read temperature (device ID varies)
cat /sys/bus/w1/devices/28-*/temperature
```

External Power Button (Pi 5 J2 Header)

Both sites have an external momentary power button wired to the Pi 5's **J2 power button header** (2-pin, located near the USB-C port). This is a **dedicated hardware power control** — it is NOT a GPIO pin. The power button also serves as the maintenance mode trigger (long press, 3 seconds).

Pi State	Action	Result
Off (halted)	Brief press	Powers on
Running	Brief press	Initiates clean shutdown
Running	Long press (3s)	Enters maintenance mode
Frozen	Hold ~10 seconds	Forces power off

No software configuration is required — the J2 header is active whenever the Pi has power from the USB-C supply.

GPIO (LEDs, button)

```
# Export GPIO pin
echo 17 > /sys/class/gpio/export
```

```
# Set as output
echo out > /sys/class/gpio/gpio17/direction

# Turn on
echo 1 > /sys/class/gpio/gpio17/value

# Turn off
echo 0 > /sys/class/gpio/gpio17/value
```

Relay Control (Electronics-Salon 4-channel SPDT, GPIO-triggered)

The relay module uses **active-high logic** (verified empirically 2026-03-26): driving GPIO HIGH energizes the relay coil (closing the NO contact). The PoE switch is wired through the NO (normally open) contact, so the relay must be energized for cameras to have power. At boot (GPIO unconfigured), the relay defaults to de-energized = cameras off (fail-safe — loads turn off safely when power is lost).

> **Note:** ORC's `orc-gpio-relays.py` uses active-low convention. A PR will be submitted to make ORC's relay polarity configurable.

```
# Relay channels driven via G469 breakout:
#   GPIO 24 → IN1 (PoE switch)   — active-high (HIGH = relay ON)
#   GPIO 18 → WS2812B NeoPixel status LED (RGB, accent via light pipe/window)

# Check relay GPIO wiring: VCC, GND, IN1-IN4 connections

# Preferred: use poe-relay utility
poe-relay on       # GPIO 24 HIGH → relay energized → PoE switch ON
poe-relay off      # GPIO 24 LOW → relay de-energized → PoE switch OFF
poe-relay status   # show current state + eth0 IP

# Alternative using pinctrl (Pi 5):
pinctrl set 24 op dh   # drive HIGH (relay ON — energized)
pinctrl set 24 op dl   # drive LOW (relay OFF — de-energized)

# Manually cycle PoE switch power
pinctrl set 24 op dl && sleep 5 && pinctrl set 24 op dh
```

Witty Pi 5 HAT+ RTC (CR2032 Coin Cell)

```
# Both sites use the Witty Pi 5 HAT+ for RTC and wake/sleep scheduling.
# The Witty Pi 5 communicates via I2C at address 0x51.
# It uses a standard CR2032 coin cell (non-rechargeable).

# Verify Witty Pi 5 is detected on I2C bus (should show 0x51)
i2cdetect -y 1

# Check hardware clock (RTC) time
sudo hwclock --show
```

```
# Sync system time to RTC
sudo hwclock --systohc

# Sync RTC time to system
sudo hwclock --hctosys

# Check Witty Pi 5 schedule and status
# (refer to Witty Pi 5 documentation for schedule script commands)

# If RTC loses time: replace the CR2032 coin cell in the Witty Pi 5 battery holder.
# CR2032 is a standard non-rechargeable cell — no config.txt charging settings needed.
```

Note: The Pi 5 built-in RTC (J5 ML-2020 connector) is NOT used. The ML-2020 battery connector failed on both Sukabumi and Jakarta boards. The Witty Pi 5 HAT+ replaces this function with a more robust CR2032 coin cell holder.

Logs

```
# System log
journalctl -xe

# ORC application log
tail -f /var/log/orc/orc.log

# Boot messages
dmesg | tail -50

# Kernel messages (USB issues)
dmesg | grep -i usb
```

Maintenance Mode

Entering Maintenance Mode

1. **Long press** power button (3+ seconds)
2. Wait for **cyan LED** steady on
3. System will:
   - Prevent auto-shutdown
   - Enable SSH access

Connecting to Maintenance Mode

1. Connect to the station via Pangolin or Tailscale:
   - Jakarta: `https://arc-00001.openrivercam.com`
   - Sukabumi: `https://arc-00002.openrivercam.com`
2. SSH into Pi via Pangolin or Tailscale
3. Default credentials:

- User: `pi`
- Password: `[configured password]`

Exiting Maintenance Mode

- **Short press** power button, OR
- Reboot: `sudo reboot`, OR
- Wait 30 minutes (auto-timeout)

When to Escalate

Contact technical support if:

1. **Hardware failure** - Component needs replacement
2. **Software bug** - ORC application error not in logs
3. **Network config** - Cannot establish LTE connection
4. **Physical damage** - Enclosure, pole, or cable damage
5. **Repeated failures** - Same component fails multiple times

Before escalating, collect: - Site name and location - Description of problem - Status LED pattern - Relevant log excerpts - Photos if physical damage - Last known working date

---

Preventive Maintenance Schedule

Monthly

- ☐ Visual inspection of enclosure (no damage, sealed)
- ☐ Check status LEDs functioning
- ☐ Verify data uploads occurring
- ☐ Check antenna connection (finger-tight on puck mount)

Quarterly

- ☐ Clean camera lens (if accessible)
- ☐ Check all bulkhead and gland seals
- ☐ Verify Gore vents not blocked
- ☐ Check rain gauge (clear any debris from RG-15 lens)
- ☐ Test maintenance mode access
- ☐ Review system logs for warnings

Annually

- ☐ Inspect conformal coating (a thin protective layer on circuit boards) on PCBs
- ☐ Check battery health (Jakarta: UPS, Sukabumi: solar)
- ☐ Replace degraded cable ties
- ☐ Test full system shutdown and restart
- ☐ Verify ground resistance $<25\Omega$ (Jakarta)
- ☐ Update OS and ORC software if available

Document Version: 3.2 **Last Updated:** April 3, 2026 **Changes from v3.1:** - Reinstated Witty Pi 5 HAT+ for both sites — Pi 5 ML-2020 battery connector (J5) broke on both boards - Replaced "Rainbow Sensing RTC (Pi 5 Built-in RTC)" section with "Witty Pi 5 HAT+ RTC (CR2032 Coin Cell)" - Removed ML-2020 battery voltage and charging config.txt references - Added Witty Pi 5 I2C address (0x51) to I2C command reference - Added note explaining why Pi 5 built-in RTC is not used

Changes from v3.0: - Added external power button (Pi 5 J2 header) to "System Won't Power On" flowchart as first diagnostic step - Added "Pi Frozen / Unresponsive" flowchart with J2 power button recovery steps - Added J2 power button rows to Power Issues table (halted Pi, frozen Pi) - Added "External Power Button (Pi 5 J2 Header)" reference section in Command Reference

Changes from v2.0: - Removed Witty Pi references from both sites; scheduling handled by Pi 5 built-in RTC (ML-2020 coin cell) via Rainbow Sensing image (subsequently reinstated in v3.2) - Replaced "Witty Pi (Sukabumi Only)" section with "Rainbow Sensing RTC (Pi 5 Built-in RTC)" (subsequently replaced in v3.2) - Updated relay control from USB-powered to GPIO-triggered (Electronics-Salon 4-ch SPDT via G469 breakout: GPIO 24→IN1, GPIO 17→IN2, GPIO 27→IN3, GPIO 22→IN4) - Replaced all "Check relay USB cable" references with "Check relay GPIO wiring (VCC, GND, IN1-IN4)" - Updated power-on flow diagram: removed Witty Pi box, DDR-60G-5 powers Pi 5 directly at both sites - Fixed DS18B20 temperature probe GPIO: was GPIO 24, corrected to GPIO 4

Changes from v1.0 (historical, some entries no longer accurate): - Removed "Check BatteryProtect setting" references (BMS handles cutoff) - Replaced "PoE injector" with "PoE switch" throughout - Replaced I2C rain gauge diagnostics with UART diagnostics for Hydreon RG-15 - Replaced "IR Light Not Working (Sukabumi Only)" with "Camera IR" section for built-in IR (both sites) - Added UART, 1-Wire, and I2C sensor command references - Updated bulkhead/gland references throughout